

RoundFi | Behavioral Reputation Score

Portable on-chain reputation for behavioral credit

Updated institutional revision after PRs 400–405 and operational hardening

1. Executive Summary

Most DeFi lending systems rely on overcollateralization because they do not have a native way to measure recurring financial discipline.

Wallet balances can show capital. Transaction history can show activity. But neither necessarily proves that a user can maintain recurring financial commitments over time.

RoundFi introduces a different signal: recurring financial behavior under commitment.

Payments → Reputation → Potentially Lower Collateral

RoundFi turns recurring payment discipline into portable on-chain behavioral reputation that may progressively support lower collateral requirements under bounded conditions.

The long-term goal is to make this reputation portable across Solana protocols and financial applications.

2. The Problem With Current On-Chain Credit Signals

DeFi credit is still heavily wealth-based. Most lending protocols focus on collateral, assets held, liquidation thresholds, wallet history and prior protocol interactions.

These are useful signals, but incomplete. A wallet with a high balance may still be a poor borrower, while a wallet with low assets may still represent a disciplined payer.

A single transaction does not prove trust. A wallet balance does not prove discipline.

3. What the RoundFi Score Measures

The RoundFi score measures behavioral financial reliability. It is based on whether a user completes recurring payment obligations inside the protocol.

- Consistency
- Punctuality
- Completed cycles
- Defaults
- Late payments

- Participation across groups
- Long-term behavioral history

The protocol assumes behavioral reliability emerges from repeated financial continuity over time rather than isolated wallet events.

4. How Reputation Is Generated

RoundFi uses savings groups as a behavioral underwriting engine. The mechanism is intentionally simple:

1. A user joins a group.
2. The user commits to recurring payments.
3. Payments are tracked across cycles.
4. Completed cycles create behavioral history.
5. This history improves the user's reputation.
6. Stronger behavioral history may justify progressively lower modeled collateral requirements under bounded economic assumptions.

A user does not buy reputation. A user earns it by completing financial commitments.

5. The Reputation Ladder

RoundFi maps reputation into a collateral ladder. The current narrative model is:

- Level 1 — Beginner — 50% stake requirement
- Level 2 — Proven — 30% stake requirement
- Level 3 — Veteran — 10% stake requirement

This ladder creates the protocol's core incentive structure: stronger behavioral history may justify progressively lower modeled collateral requirements under bounded assumptions.

The current collateral-reduction model remains under empirical behavioral validation and should not yet be interpreted as production-grade undercollateralized credit.

6. Why This Matters to Users

Users do not join RoundFi because they want an abstract score. Users join because a stronger behavioral profile may eventually reduce how much capital they need to lock.

- Lower modeled collateral requirements
- Better potential access to credit
- More capital efficiency
- Reusable financial reputation
- Path toward future undercollateralized experimentation

The protocol still requires real-world validation regarding how effectively behavioral reputation correlates with lower financial risk.

7. Why This Matters to Protocols

Protocols need better underwriting signals. A portable behavioral score can help evaluate whether a wallet has completed recurring commitments, paid on time, defaulted or built a consistent behavioral history over time.

This may eventually become useful for lending protocols, credit applications, undercollateralized systems, fintechs, neobanks, risk engines and B2B underwriting infrastructure.

The groups generate the data. The reputation layer monetizes access to behavioral coordination signals.

8. Attestations and Portability

RoundFi's reputation thesis depends on portability. If reputation only works inside one application, its usefulness remains limited.

The goal is to make behavioral reputation readable and reusable across protocols. RoundFi is designed around SAS-compatible attestations and a reputation program capable of exposing behavioral profiles to external applications.

RoundFi currently frames portability as progressively evolving behavioral infrastructure rather than finalized production-grade underwriting infrastructure.

9. Score Inputs

A behavioral credit score should not depend on one isolated event. RoundFi's score concept is built around repeated events over time.

Input signals include:

- on-time payments;
- completed cycles;
- late payments;
- defaults;
- participation continuity;
- optional identity hints.

The protocol intentionally treats completed recurring financial behavior as the primary source of reputation generation.

10. Anti-Gaming Principles

Any reputation system can be attacked if users can farm reputation cheaply. RoundFi therefore attempts to make reputation economically expensive to fake.

- recurring commitments;
- capital at risk;
- cycle completion requirements;
- default stickiness;
- cooldowns and rate limits;
- optional identity hints.

RoundFi increasingly treats anti-gaming as an economic-friction problem rather than purely an identity problem. The protocol attempts to make reputation expensive to manufacture through persistence requirements, recurring obligations and long-duration behavioral continuity.

11. Current Implementation Status

RoundFi currently remains in hackathon-stage validation with a meaningful technical foundation.

The repository currently demonstrates:

- functional prototype;
- live front-end experience;
- reputation page and score interface;
- Stress Lab simulations;
- Solana Devnet deployment;
- open-source codebase;
- app-to-chain flows;
- reputation program architecture;
- operational hardening infrastructure.

Recent protocol evolution also introduced stronger regression philosophy, canary-stage operational posture and adversarial replay awareness following operational hardening cycles.

12. Why the Score Is the Product

The savings group is the mechanism. The score is the asset.

The group generates recurring behavior. The behavior creates reputation. The reputation may eventually become reusable across financial applications.

If RoundFi is positioned only as a Web3 savings group, the opportunity appears smaller and easier to copy. If RoundFi is positioned as behavioral reputation infrastructure, the strategic opportunity becomes significantly larger.

13. Business Model Connection

The reputation score connects directly to the RoundFi business model.

- participation fees;
- infrastructure integrations;
- future reputation APIs;
- underwriting infrastructure services.

The long-term strategic opportunity remains the infrastructure layer rather than the savings group itself.

14. Key Risks

RoundFi still faces meaningful uncertainty.

Important unresolved risks include:

- score acceptance risk;
- gaming risk;
- regulatory risk;
- data quality risk;
- integration risk;
- UX risk;
- behavioral validation risk.

Real-world users may behave differently under meaningful financial stress than modeled conditions predict.

15. What RoundFi Must Prove Next

RoundFi still needs to prove several core assumptions empirically:

1. Users are willing to complete recurring payment cycles.
2. The score can resist cheap farming.
3. Behavioral incentives meaningfully influence user behavior.
4. Coordination durability remains stable under pressure.
5. Collateral reduction can safely correlate with behavioral history.
6. Protocols are willing to consume the score.
7. The B2B API has real demand.
8. Reputation portability remains economically useful.
9. Long-term retention quality remains healthy.
10. The system can safely progress toward controlled mainnet deployment.

16. One-Sentence Summary

RoundFi's Behavioral Reputation Score turns recurring payment discipline into portable on-chain behavioral reputation that may progressively support lower collateral requirements under bounded conditions.

17. Related Documents

- [01-roundfi-overview.md](#)
- [02-technical-whitepaper.md](#)
- [03-architecture-spec.md](#)
- [05-stress-lab-economic-model.md](#)
- [06-market-and-gtm.md](#)
- [07-business-model-b2b-oracle.md](#)
- [08-competitive-analysis.md](#)
- [09-risk-and-compliance.md](#)
- [11-devnet-status-and-proof.md](#)

18. References

- [RoundFi GitHub README](#)
- [RoundFi Project Status Register](#)
- [RoundFi Architecture Specification](#)
- [RoundFi Stress Lab documentation](#)
- [Solana Attestation Service documentation](#)
- [Solana developer documentation](#)
- [Kamino Finance documentation](#)
- [Switchboard documentation](#)